



OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS IN
ITP422 – NETWORKING 1
Date enhanced: August 2026

Academic Year/ Semester	AY 2026-2027 / 1 st Semester		
University	Vision: A globally-recognized green University committed to excellent service to the nation. Mission: Foster excellence, holistic and inclusive education attuned to the demands of a diverse world towards the formation of productive and value-laden lives. Goal: Uphold the tradition of excellence in instruction, research, extension and production geared towards a strong academe-industry interface for societal development. Core Values: Respect, Integrity, Social Responsibility, Excellence, Commitment Institutional Outcomes: 1. A <i>Professional</i> who is morally upright, socially responsible and globally employable 2. A <i>Leader and Innovator</i> who inspires others and is committed to social transformation and national development 3. An <i>Environmental Advocate</i> committed to research, extension and production initiatives		
Course Title	ITP422 – NETWORKING 1		
Course Credits	3 units (2 units lecture, 1 unit laboratory)		
Course Description	This course emphasizes the knowledge and application of basic concepts of networking technology. It presents the OSI model, industry standards, network topologies, IP addressing, subnet masking, networking components, routing protocols, and basic network design with laboratory experience.		
SDG Focus	SDG 9 – Industry, Innovation and Infrastructure		
Contact Hours/Week	5 hours / week (2 hours lecture, 3 hours laboratory)		
Days/Time	BSIT 2A: MTH / 9:30AM-12:00PM BSIS 2B: TF / 7:00AM-12:00PM		
Pre-requisite Course Code- Course Title	CC101, ITRACKA1		
Program Outcomes	F. Apply knowledge of computing science, and mathematics appropriate to the discipline.	Program Indicators	Solve algorithmic problems and apply mathematical models
	G. Understand best practices and standards and their applications.		Implement coding standards and IT frameworks



		H. Analyze complex problems and identify and define the computing requirements appropriate to its solution.		Conduct problem analysis and requirements gathering
		J. Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.		Develop prototypes and evaluate against requirements
		K. Integrate IT-based solutions into the user environment effectively.		Deploy IT systems and provide training/documentation
		L. Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.		Demonstrate proficiency in programming languages and IT tools
		Q. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.		Evaluate IT case studies on privacy, IP, cybersecurity
Course Outcomes		Describe the data communications and network models, topologies, protocols, standards and architectures.	Course Indicators	Demonstrate understanding of the fundamental concepts, models, standards, protocols, and architectures of data communications and computer networks.
		Describe necessary hardware and components used to establish communication between multiple networks and analyses the effect of various topologies, applications and devices on network performance.		Determine the ability to identify networking hardware and components and analyze their roles and effects on network performance.
		Analyze routing algorithms protocols, process routing tables, and configure routers for proper orientation of an efficient network		Evaluates routing mechanisms and configure routers to establish efficient and reliable network communication.
Course Outline and Timeframe				
Week 1-3	15 hrs	Course Orientation A – Networking Basics		
Week 4-9	27.5 hrs	B – Networking Protocols		
Week 9	2.5 hrs	MIDTERM EXAMINATION		
Week 10-13	20 hrs	C – TCP/IP Services		
Week 14-18	22.5 hrs	D- Network Management / Troubleshooting		
Week 18	2.5 hrs	FINAL EXAMINATION		
Alignment of Course Outcomes with Summary Assessment Tasks				
Course Objectives		Summative Assessment Task	Details	
Explain the fundamental concepts of data communications, network models, topologies, protocols, standards, and network architectures.		Written Examination and Concept Mapping	Students complete a written examination covering networking concepts, models, protocols, standards, and architectures, and prepare a concept map illustrating the relationships among networking components and communication processes.	



Identify and analyze networking hardware, devices, and components used in establishing communication between multiple networks and evaluate their impact on network performance.	Network Design and Laboratory Performance Task	Students identify appropriate networking devices and media for a given scenario, design a basic network topology using a network simulation tool (e.g., Cisco Packet Tracer), and analyze how the selected hardware, topology, and devices affect network performance.
Apply routing concepts by analyzing routing algorithms and protocols, interpreting routing tables, and configuring routers to establish efficient and reliable network communication.	Router Configuration Practical Examination	Students configure routers using static and/or dynamic routing protocols in a simulated network environment, interpret routing tables, verify end-to-end connectivity, and troubleshoot routing issues to achieve efficient network communication.

Learning Plan

<i>Desired Learning Outcomes</i>	<i>Content Area</i>	<i>SDGs</i>	<i>Resources/ Materials</i>	<i>Teaching and Learning Activities*</i>	<i>Assessment Tasks</i>	<i>References</i>	<i>Time Frame and Contact Hours.</i>
At the end of the lesson the students shall be able to: 1. Explain the basics terminologies used in networking. 2. Differentiate data communication systems and its components. 3. Create a simple network	A – Networking Basics 1. Introduction to Networking 2. Network Infrastructure 3. -Transmission & Hardware 4. Switches, Routers, and VLANs 5. Servers and Virtualization 6. Cloud Computing	SDG 9 – Industry, Innovation and Infrastructure	- Laptop/TV/Cellphones - PowerPoint presentation - Internet Connectivity	- Lecture - Interactive discussion - Brainstorming - Individual/Group Activity	- Graded Recitation - Quizzes - Assignment / Seatwork - Activities	[1] White, G., et.al. (2021). Principles of computer security: CompTIA Security+ and beyond (Exam SY0-601) (6th ed.). McGraw-Hill Education. [2] Cisco Networking Academy(2024) . Networking Essentials Companion Guide v3: Cisco Certified Support Technician (CCST) Networking 100-150. Cisco Press. ISBN: 9780138321352, 9780137660483, 0136633668.	Week 1-3 (15 hrs)
1. Explain the Network Theory.	B – Networking Protocols 1. Protocol Models	SDG 9 – Industry, Innovation	Laptop/TV/Cellphones	- Lecture - Interactive discussion	- Graded Recitation - Quizzes	Ferrel, Shadow, Akamai Technologies, Inc.	Week 4-9 (27.5hrs)



2. Explain the categories of network and the standard network model.	2. Protocols and Standards 3. IPV4 and IPV6 4. IP Addresses 5. Subnetting	and Infrastructure	- PowerPoint presentation - Internet Connectivity - Packet Tracer	- Brainstorming - Individual/Group Activity	- Assignment / Seatwork - Activities	https://www.coursera.org/learn/akamai-networking#modules	
3. Implement Cisco packet tracer							
MIDTERM EXAMINATION							Week 9 2.5 hrs
1. Explain the DHCP services that support TCP/IP.	C – TCP/IP Services 1. Static and Dynamic Addressing 2. The DHCP Lease Process 3. Centralized DHCP 4. DHCP Server Settings 5. DNS Overview	SDG 9 – Industry, Innovation and Infrastructure	- Laptop/TV/Cellphones - PowerPoint presentation - Internet Connectivity - Packet Tracer	- Lecture - Interactive discussion - Brainstorming - Individual/Group Activity	- Graded Recitation - Quizzes - Assignment / Seatwork - Activities	Ferrel, Shadow, Akamai Technologies, Inc. https://www.coursera.org/learn/akamai-networking#modules	Week 10-13 (20 hrs)
2. Identify what is DHCP.							
1. Explain the concept of network management	D- Network Management / Troubleshooting 1. Network Management 2. Different Issue of Network 1. Physical Issues 2. Logical Issues 3. Wireless Issues 3. Troubleshooting hint 1. Physical/Logical Physical Issues 2. Wireless Issues	SDG 9 – Industry, Innovation and Infrastructure	- Laptop/TV/Cellphones - PowerPoint presentation - Internet Connectivity - Packet Tracer	- Lecture - Interactive discussion - Brainstorming - Individual/Group Activity	- Graded Recitation - Quizzes - Assignment / Seatwork - Activities	Ferrel, Shadow, Akamai Technologies, Inc. https://www.coursera.org/learn/akamai-networking#modules	Week 14-18 (22.5hrs)
2. Learn the troubleshooting methodology and review strategies for troubleshooting various type of issues.							
FINAL EXAMINATION							Week 18 2.5 hrs

* Flexible learning modalities shall be implemented in the event of suspended face-to-face classes due to weather-related emergencies or other institutional interventions.



Suggested Additional References:

1. Ranjan, P. et.al. (2023). Wireless Communications and Networking Technologies: Classifications, Advancement and Applications. CRC Press. ISBN: 9781003181699 (ebk)
2. Zhang, Y. (2018). Network function virtualization: Concepts and applicability in 5G networks. John Wiley & Sons.
3. Liyanage, M., et.al. (2018). A comprehensive guide to 5G security. John Wiley & Sons.
4. Lowe, Doug,(2018). Networking All-in-One, 7th edition, John Wiley & Sons, Inc., NJ 07030-5774
5. Ferrell, Shadow Networking Fundamental <https://www.coursera.org/learn/akamai-networking#modules>

Course Requirements:

1. Attendance and participation in the class discussion
2. Group or Individual activity/group outputs
3. Quizzes/Seatwork
4. Mid-term and final examination

Grading System:

(Per Board Resolution No. 3, 4 or 5, s. 2016)

The final grade of each student will be computed as follows:

Midterm/Final Examination	30%
Quizzes/Recitation/Project/Problem Sets	30%
Performance (skill based; psychomotor)	40%
TOTAL	100%

Distribution of Weight for the Overall/Final Grade

Mid-Term Grade	50%
Tentative	50%

Course Assessment Measures:

Written examinations, quizzes, concept maps, and oral recitation assessing understanding of networking concepts and principles. Laboratory exercises, network design activities, simulation-based tasks (e.g., Cisco Packet Tracer), and performance rubrics. Practical laboratory examination, router configuration tasks, troubleshooting exercises, and project-based assessments using network simulation tools.

Continuous Quality Improvement Measures:

Review students' assessment results regularly to identify areas for improvement, update learning materials and laboratory activities as needed, and provide additional practice and support to help students strengthen their networking knowledge and practical skills.

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